

PIL2-STARK: Commitment Phase

Prover		Verifier
<i>Stage 1: Witness</i>		
Hold witness polynomials $f_1, \dots, f_m$		
Extend each f_j to H^* , commit:	$\longrightarrow$	r_1
$r_1 = \text{MT}(f_1, \dots, f_m)$		Seed transcript $\mathcal{T}$ from $(\text{vk}, \text{pub}, r_1)$
<i>Stage 2: Intermediate polynomials</i>		
$(\alpha, \gamma, \dots)$	$\longleftarrow$	$\alpha, \gamma, \dots \leftarrow \mathcal{T}.\text{squeeze}()$
Compute intermediate columns $h_1, \dots, h_k$ using challenges α, γ		
Extend each h_j to H^* , commit:	$\longrightarrow$	r_2
$r_2 = \text{MT}(h_1, \dots, h_k)$		$\mathcal{T}.\text{absorb}(r_2)$
<i>Stage Q: Quotient polynomial</i>		
v_c	$\longleftarrow$	$v_c \leftarrow \mathcal{T}.\text{squeeze}()$
Evaluate combined constraint on H^* :		
$C(x) = \sum_{j=0}^{J-1} v_c^{J-1-j} C_j(x)$		
Compute $Q(x) = C(x)/Z_H(x)$		
Split: $Q = Q_0 + X^N Q_1 + \dots + X^{(d-1)N} Q_{d-1}$		
Commit: $r_Q = \text{MT}(Q_0, \dots, Q_{d-1})$	$\longrightarrow$	r_Q $\mathcal{T}.\text{absorb}(r_Q)$
<i>Evaluation stage</i>		
ξ	$\longleftarrow$	$\xi \leftarrow \mathcal{T}.\text{squeeze}()$
For each polynomial p and offset $o \in \mathcal{O}$:		
$e_{p,o} = p(\xi \cdot \omega^o) \in \mathbb{F}_{p^3}$		
$\{e_{p,o}\}$	$\longrightarrow$	$\{e_{p,o}\}$ $\mathcal{T}.\text{absorb}(\text{LinearHash}(\{e_{p,o}\}))$
<i>FRI polynomial</i>		
(v_1, v_2)	$\longleftarrow$	$v_1, v_2 \leftarrow \mathcal{T}.\text{squeeze}()$
Compute FRI polynomial F on H^* :		
$F(x) = \sum_g v_1^{ G -1-g} \cdot \frac{\sum_j v_2^{n_g-j} (p_j(x) - e_j)}{x - \xi \omega^{o_g}}$		
<i>FRI commitment rounds</i>		
Set $F_0 = F$		
Commit: $r_0^{\text{FRI}} = \text{MT}(F_0)$	$\longrightarrow$	r_0^{FRI} $\mathcal{T}.\text{absorb}(r_0^{\text{FRI}})$
β_0	$\longleftarrow$	$\beta_0 \leftarrow \mathcal{T}.\text{squeeze}()$
Fold: $F_1 = \text{Fold}(F_0, \beta_0)$		
$\vdots$		$\vdots$
Commit: $r_{K-1}^{\text{FRI}} = \text{MT}(F_{K-1})$	$\longrightarrow$	r_{K-1}^{FRI} $\mathcal{T}.\text{absorb}(r_{K-1}^{\text{FRI}})$
β_{K-1}	$\longleftarrow$	$\beta_{K-1} \leftarrow \mathcal{T}.\text{squeeze}()$
Fold: $F_K = \text{Fold}(F_{K-1}, \beta_{K-1})$		
F_K	$\longrightarrow$	F_K $\mathcal{T}.\text{absorb}(\text{LinearHash}(F_K))$ Check: $\deg(F_K) < D$
<i>Proof of work</i>		
χ_{grind}	$\longleftarrow$	$\chi_{\text{grind}} \leftarrow \mathcal{T}.\text{squeeze}()$
Find η such that $\text{Poseidon2}(\chi_{\text{grind}} \parallel \eta)$ has b_{pow} leading zero bits		
η	$\longrightarrow$	η Check: $\text{Poseidon2}(\chi_{\text{grind}} \parallel \eta)$ has b_{pow} leading zeros